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BATTERY AND BATTERY MODULE

Abstract

In an embodiment, a battery includes an exterior part, an electrode group, a terminal, and an insulator, and the electrode group is housed in an internal cavity of the exterior part. The terminal is inserted into the internal cavity through the exterior part and includes a terminal head arranged outside the exterior part. The insulator includes a bottom portion sandwiched between the terminal head and an outer surface of the exterior part, and a covering portion covering the terminal head from an outer peripheral side of the terminal on the outer surface of the exterior part. The covering portion abuts on the outer surface of the exterior part, and a protruding height of the covering portion from the outer surface of the exterior part is a protruding height of the terminal head from the outer surface of the exterior part or more over an entire circumference.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a Continuation application of PCT Application No. PCT/JP2023/010108, filed Mar. 15, 2023, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments relate to a battery and a battery module.

BACKGROUND

[0003] As a battery such as a lithium ion secondary battery, there is a battery in which an electrode group is housed in an internal cavity of an exterior part. In such a battery, a terminal (electrode terminal) is attached to the exterior part, and the terminal is electrically connected to the electrode group via a lead or the like. In the battery, an electrical path from the electrode group is formed through the lead and the terminal. As terminals used in batteries, a terminal including a terminal head is available. In the battery in which the terminal with a terminal head is used, the terminal is inserted through the exterior part into the internal cavity, and the terminal head is arranged outside the exterior part.

[0004] In the case where a battery module in which a plurality of batteries are electrically connected is assembled using the above-described batteries, bus bars are connected to terminals of the batteries, and the plurality of batteries are electrically connected via the bus bars. In a work of assembling the battery module, to confirm the dimensions of the battery module to be assembled, measurement and the like of the dimensions of the batteries forming the battery module are performed. In work such as assembling a battery module using a battery in which a terminal head of a terminal is arranged outside an exterior part, it is required to appropriately prevent unintended contact of the terminal head arranged outside the exterior part with a conductor so as to appropriately prevent occurrence of a short circuit caused by contact of the terminal head with the conductor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a perspective view showing a battery according to a first embodiment.

[0006] FIG. 2 is a schematic view showing the battery according to the first embodiment as viewed from one side in a depth direction.

[0007] FIG. 3 is a plan view showing the battery according to the first embodiment as viewed from one side in a height direction.

[0008] FIG. 4 is a cross-sectional view showing the battery according to the first embodiment, taken along a cross-section orthogonal or substantially orthogonal to a lateral direction.

[0009] FIG. 5 is a cross-sectional view showing a configuration of one of a pair of terminals and a vicinity thereof in the battery according to the first embodiment, which is taken along a cross section orthogonal or substantially orthogonal to the lateral direction of the battery.

[0010] FIG. 6 is a perspective view showing a battery module according to the first embodiment.

[0011] FIG. 7 is a cross-sectional view explaining a configuration in which bus bars are connected to terminals of batteries in the battery module according to the first embodiment.

[0012] FIG. 8 is a perspective view showing a battery module according to a first modification.

[0013] FIG. 9 is a perspective view showing a battery module according to a second modification.

DETAILED DESCRIPTION

[0014] According to an embodiment, a battery includes an exterior part, an electrode group, a terminal, and an insulator, and the exterior part has an internal cavity. The electrode group includes a positive electrode and a negative electrode and is housed in the internal cavity. The terminal is inserted into the internal cavity through the exterior part and is electrically connected to the electrode group. The terminal includes a terminal head arranged outside the exterior part. The insulator includes a bottom portion sandwiched between the terminal head and the outer surface of the exterior part, and a covering portion covering the terminal head from an outer peripheral side of the terminal on the outer surface of the exterior part, and has electrical insulation properties. The covering portion abuts on the outer surface of the exterior part, and the protruding height of the covering portion from the outer surface of the exterior part is the protruding height of the terminal head from the outer surface of the exterior part or more over the entire circumference.

[0015] Hereinafter, embodiments will be described with reference to the drawings.

First Embodiment

[0016] Hereinafter, a first embodiment will be described as an example of the embodiment. First, a single battery **1**, which is a unit cell used in the first embodiment, will be described. FIGS. **1**, **2**, **3**, and **4** illustrate the battery **1** according to the first embodiment. As shown in FIGS. **1** to **4**, the battery **1** includes an electrode group **2** and an exterior part **3**. In the example in FIGS. **1** to **4**, the exterior part **3** includes an exterior container **5** and a lid member **6**. The exterior part **3** including the exterior container **5** and the lid member **6** is formed of a metal such as aluminum, an aluminum alloy, iron, copper, or stainless steel, and has conductivity.

[0017] In the example of the battery **1** (exterior part **3**) in FIGS. **1** to **4**, a depth direction (a direction indicated by arrows **X1** and **X2**), a lateral direction (a direction indicated by arrows **Y1** and **Y2**) intersecting (orthogonal or substantially orthogonal to) the depth direction, and a height direction (a direction indicated by arrows **Z1** and **Z2**) intersecting (orthogonal or substantially orthogonal to) both the depth direction and the lateral direction are defined. In each of the battery **1** and the exterior part **3**, the dimension in the depth direction is smaller than each of the dimension in the lateral direction and the dimension in the height direction. FIG. **1** is a perspective view, and FIG. **2** is a view as viewed from one side in the depth direction. FIG. **3** is a view as viewed from one side in the height direction, and FIG. **4** shows a cross section orthogonal or substantially orthogonal to the lateral direction.

[0018] The exterior container **5** includes a bottom wall **7** and a peripheral wall **8**. An internal cavity **10** is formed inside the exterior part **3**, and the internal cavity **10** is defined by the bottom wall **7** and the peripheral wall **8** of the exterior container **5**. In the exterior container **5**, the internal cavity **10** is open toward the side opposite to the side where the bottom wall **7** is located in the height direction. The peripheral wall **8** includes two pairs of side walls **11** and **12**. The pair of side walls **11** are opposed to each other in the lateral direction with the internal cavity **10** therebetween. The pair of side walls **12** are opposed to each other in the depth direction with the internal cavity **10** therebetween. Each of the side walls **11** is continuously extended along the depth direction between the side walls **12**. Each of the side walls **12** extends continuously along the lateral direction between the side walls **11**. The lid member **6** is attached to the peripheral wall **8** to an end portion in the opposite side to the bottom wall **7**. Thus, the lid member **6** closes the opening of the internal cavity **10**. The lid member **6** and the bottom wall **7** are opposed to each other in the height direction with the internal cavity **10** therebetween. The lid member **6** is attached to the peripheral wall **8** by welding such as seam welding.

[0019] In the example in FIGS. **1** to **4**, the lid member **6** is attached to the peripheral wall **8** of the exterior container **5** in a state where the thickness direction of the lid member **6** coincides or substantially coincides with the height direction of the battery **1**. The lid member **6** includes a

peripheral edge **14** forming an edge of the outer surface. In the battery **1**, the peripheral edge **14** is welded to the peripheral wall **8** of the exterior container **5**. The peripheral edge **14** includes a pair of short edges **17** and a pair of long edges **18**. The pair of short edges **17** are positioned apart from each other in the lateral direction of the battery **1**, and each of the short edges **17** is continuously extended along the depth direction of the battery **1** between the pair of long edges **18**. The pair of long edges **18** are positioned apart from each other in the depth direction of the battery **1**, and each of the long edges **18** is continuously extended along the lateral direction of the battery **1** between the pair of short edges **17**. In the battery **1**, each of the short edges **17** is welded to a corresponding one of the pair of side walls **11**, and each of the long edges **18** is welded to a corresponding one of the pair of side walls **12**. The dimension of the lid member **6** in the depth direction of the battery **1** (a direction along the short edge **17**) is smaller than the dimension of the lid member **6** in the lateral direction of the battery **1** (a direction along the long edge **18**).

[0020] The electrode group **2** is housed in the internal cavity **10**. In FIG. **2**, some of the components housed in the internal cavity **10**, such as the electrode group **2**, are illustrated by broken lines. The electrode group **2** includes a positive electrode **13A** and a negative electrode **13B**. In the electrode group **2**, a separator (not shown) is interposed between the positive electrode **13A** and the negative electrode **13B**. The separator has electrical insulation properties and electrically insulates the positive electrode **13A** from the negative electrode **13B**.

[0021] The positive electrode **13A** includes a positive electrode current collector such as a positive electrode current collector foil and a positive electrode active material-containing layer (not shown) supported on a surface of the positive electrode current collector. The positive electrode current collector is, but is not limited to, for example, an aluminum foil or an aluminum alloy foil, and has a thickness of approximately 10 μm to 20 μm . The positive electrode active material-containing layer includes a positive electrode active material, and may optionally include a binder and an electro-conductive agent. Examples of the positive electrode active material include, but are not limited to, oxides, sulfides, polymers, and the like capable of occluding and releasing lithium ions. The positive electrode current collector includes a positive electrode current collecting tab **15A** as a portion not supporting the positive electrode active material-containing layer.

[0022] The negative electrode **13B** includes a negative electrode current collector such as a negative electrode current collector foil, and a negative electrode active material-containing layer (not shown) supported on a surface of the negative electrode current collector. The negative electrode current collector is, but is not limited to, for example, an aluminum foil, an aluminum alloy foil, or a copper foil, and has a thickness of approximately 10 μm to 20 μm . The negative electrode active material-containing layer includes a negative electrode active material, and may optionally include a binder and an electro-conductive agent. The negative electrode active material is not particularly limited, and examples thereof include metal oxides, metal sulfides, metal nitrides, and carbon materials capable of occluding and releasing lithium ions. The negative electrode current collector includes a negative electrode current collecting tab **15B** as a portion not supporting the negative electrode active material-containing layer.

[0023] In the example of the electrode group **2** shown in FIGS. **2** and **3**, the positive electrode **13A**, the negative electrode **13B**, and the separator are wound in a state where the separator is sandwiched between the negative electrode active material-containing layer and the positive electrode active material-containing layer, thus forming the electrode group **2** having a wound structure. In another example, the electrode group **2** has a stack structure in which a plurality of positive electrode plates serving as a positive electrode **13A** and a plurality of negative electrode plates serving as a negative electrode **13B** are alternately stacked, and a separator is provided between the positive electrode plate and the negative electrode plate. As a result of forming the electrode group **2** as described above, in the electrode group **2**, the positive electrode current collecting tab **15A** and the negative electrode current collecting tab **15B** are formed as a pair of current collecting tabs **15**. The pair of current collecting tabs **15** do not contact each other and are

electrically insulated from each other.

[0024] In the electrode group **2**, each of the pair of current collecting tabs **15** protrudes with respect to the positive electrode active material-containing layer, the negative electrode active material-containing layer, the separator, and the like. In the example in FIGS. **2** and **3**, the pair of current collecting tabs **15** protrude to opposite sides to each other. The electrode group **2** is arranged in the internal cavity **10** in a state where the positive electrode current collecting tab **15A** protrudes to one side in the lateral direction of the battery **1** and the negative electrode current collecting tab **15B** protrudes to the side opposite to the side where the positive electrode current collecting tab **15A** protrudes in the lateral direction of the battery **1**. In an example of the electrode group **2**, the pair of current collecting tabs **15** protrude toward the same side. In this case, the electrode group **2** is arranged in the internal cavity **10** in a state where the pair of current collecting tabs **15** (the positive electrode current collecting tab **15A** and the negative electrode current collecting tab **15B**) protrude toward the side where the lid member **6** is located in the height direction of the battery **1**. The pair of current collecting tabs **15** are arranged apart from each other in the lateral direction of the battery **1**.

[0025] In the internal cavity **10**, the electrode group **2** holds (is impregnated with) an electrolytic solution (not shown). The electrolytic solution may be a nonaqueous electrolytic solution in which an electrolyte is dissolved in an organic solvent, or may be an aqueous electrolytic solution such as an aqueous solution. Instead of the electrolytic solution, a gel electrolyte may be used, or a solid electrolyte may be used. In the case where the solid electrolyte is used as the electrolyte, the solid electrolyte is interposed between the positive electrode **13A** and the negative electrode **13B** instead of the separator in the electrode group. In this case, the positive electrode **13A** is electrically insulated from the negative electrode **13B** by the solid electrolyte.

[0026] In the battery **1**, a pair of terminals **16** are attached to the lid member **6** of the exterior part **3**. The terminal **16** is formed of a conductive material such as metal. One of the pair of terminals **16** is a positive electrode terminal **16A** of the battery **1**, and the other of the pair of terminals **16** which is different from the positive electrode terminal **16A** is a negative electrode terminal **16B** of the battery **1**. Each of the terminals **16** is attached to the lid member **6** in a state where a part of the terminal **16** is located outside the exterior part **3**. Each of the terminals **16** penetrates the lid member **6** along the height direction of the battery **1**, and is inserted into the internal cavity **10** penetrating through the lid member **6**. The pair of terminals **16** are arranged apart from each other in the lateral direction of the battery **1**. In FIG. **4**, a cross section through one of the pair of terminals **16** is shown.

[0027] A pair of leads **20** are arranged in the internal cavity **10** of the exterior part **3**. In FIG. **2**, the pair of leads **20** are shown by a broken line together with the electrode group **2**. One of the pair of leads **20** is a positive electrode side lead **20A**, and the other of the pair of leads **20** which is different from the positive electrode side lead **20A** is a negative electrode side lead **20B**. Each of the leads **20** is formed of a conductive material such as metal. Examples of the conductive material forming the lead **20** include aluminum, stainless steel, copper, and iron. The positive electrode side lead **20A** forms at least a part of an electrical path between the positive electrode current collecting tab **15A** and the positive electrode terminal **16A**, and connects the positive electrode **13A** of the electrode group **2** and the positive electrode terminal **16A**. Thus, the positive electrode terminal **16A** is electrically connected to the positive electrode **13A** of the electrode group **2**. The negative electrode side lead **20B** forms at least a part of an electrical path between the negative electrode current collecting tab **15B** and the negative electrode terminal **16B**, and connects the negative electrode **13B** of the electrode group **2** and the negative electrode terminal **16B**. Thus, the negative electrode terminal **16B** is electrically connected to the negative electrode **13B** of the electrode group **2**.

[0028] As described above, in the battery **1**, each of the pair of current collecting tabs **15** is connected to a corresponding one of the pair of terminals **16** via a corresponding one of the pair of

leads **20**. The shape of each lead **20** is not limited to the shapes shown in figures such as FIGS. **2** and **4**. In the example in FIGS. **2** and **4**, each of the leads **20** includes two leg portions, but in an example, each of the leads **20** may include only one leg portion. In the configuration in which the pair of current collecting tabs **15** (the positive electrode current collecting tab **15A** and the negative electrode current collecting tab **15B**) in the electrode group **2** protrude toward the side where the lid member **6** is located in the height direction, the shape of each lead **20**, the arrangement of each lead **20** in the internal cavity **10**, and the like are different from those in the configuration of the example shown in FIGS. **2** and **4**. However, even in the configuration in which the pair of current collecting tabs **15** protrude toward the side where the lid member **6** is located in the height direction, each of the pair of current collecting tabs **15** is connected to a corresponding one of the pair of terminals **16** via a corresponding one of the pair of leads **20**.

[0029] As shown in FIG. **4**, etc., in the internal cavity **10**, an electrode group holder **21** serving as an internal insulator is arranged between the electrode group **2** and the lid member **6** in the height direction of the battery **1**. The electrode group holder **21** is arranged between each of the leads **20** and the lid member **6** in the height direction of the battery **1**. The electrode group holder **21** has electrical insulation properties. The pair of current collecting tabs **15** and the pair of leads **20** are prevented from contacting the lid member **6** by the electrode group holder **21**, and are electrically insulated from the lid member **6**. In the internal cavity **10**, each of the terminals **16** penetrates the electrode group holder **21** along the height direction of the battery **1**, and is connected to a corresponding one of the leads **20** in a state of penetrating the electrode group holder **21**.

[0030] A pair of insulating guards (not shown) are arranged as internal insulators in the internal cavity **10** of the exterior part **3**. Each of the insulating guards has electrical insulation properties. Each of the pair of insulating guards prevents the corresponding one of the pair of current collecting tabs **15** and the corresponding one of the pair of leads **20** from contacting the exterior container **5**. Thus, the pair of current collecting tabs **15** and the pair of leads **20** are electrically insulated from the exterior container **5**. Therefore, the electrode group holder **21** and the insulating guard serving as the internal insulator electrically insulate the pair of current collecting tabs **15** and the pair of leads **20** from the exterior part **3** including the exterior container **5** and the lid member **6**.

[0031] In the battery **1** of the example shown in FIGS. **1** to **4**, a gas release valve **22** and a liquid injection port (not shown) are formed in the lid member **6**. A sealing plate **23** to close the liquid injection port is welded to the outer surface of the lid member **6**. The gas release valve **22** and the liquid injection port are arranged between the pair of terminals **16** in the lateral direction of the battery **1**. In the battery **1**, the gas release valve **22**, the liquid injection port, and the like may not be provided.

[0032] In the battery **1**, the lid member **6** of the exterior part **3** is provided with a pair of through holes **25**. In the lid member **6**, the pair of through holes **25** are formed apart from each other in the lateral direction of the battery **1**, and only one of the pair of through holes **25** is shown in FIG. **4**. Each of the through holes **25** extends through the lid member **6** along the height direction of the battery **1** from the outer surface of the lid member **6** (the outer surface of the exterior part **3**) to the internal cavity **10**. Each of the pair of terminals **16** penetrates the lid member **6** in a corresponding one of the pair of through holes **25**, and is inserted into the internal cavity **10** from the corresponding one of the through holes **25**. Then, each of the pair of terminals **16** is attached to the lid member **6** to the corresponding one of the pair of through holes **25** and the vicinity thereof. One of the pair of through holes **25** into which the positive electrode terminal **16A** is inserted is a positive electrode side through hole, and one of the pair of through holes **25** into which the negative electrode terminal **16B** is inserted is a negative electrode side through hole.

[0033] The battery **1** includes a pair of insulators **26** and a pair of gaskets **27**. Each of the insulator **26** and the gasket **27** are provided corresponding to the positive electrode terminal **16A** and each of the insulator **26** and the gasket **27** are provided corresponding to the negative electrode terminal **16B**. One of the pair of insulators (terminal insulators) **26** corresponding to the positive electrode

terminal **16A** is a positive electrode side insulator **26A**, and one of the pair of insulators **26** corresponding to the negative electrode terminal **16B** is a negative electrode side insulator **26B**. One of the pair of gaskets **27** corresponding to the positive electrode terminal **16A** is a positive electrode side gasket, and one of the pair of gaskets **27** corresponding to the negative electrode terminal **16B** is a negative electrode side gasket. In FIG. **4**, only one of the pair of insulators **26** is shown, and only one of the pair of gaskets **27** is shown.

[0034] Each of the insulators (terminal insulators) **26** has electrical insulation properties, and is formed of, for example, a resin having electrical insulation properties. Each of the gaskets **27** has electrical insulation properties, and is formed of, for example, a resin having electrical insulation properties. Each of the gaskets **27** is formed in, for example, a ring shape or a substantially ring shape. In the present embodiment, each of the terminals **16** is prevented from contacting the lid member **6** by the corresponding one of the insulators **26** and the corresponding one of the gaskets **27** in the mounting portion to the lid member **6** of the exterior part **3**, and is electrically insulated from the exterior part **3** including the lid member **6**.

[0035] Hereinafter, the configuration of one of the pair of terminals **16** and the configuration of the mounting portion of the terminal **16** to the exterior part **3** (lid member **6**) will be described. FIG. **5** shows a configuration of one of the pair of terminals **16** and the vicinity thereof. FIG. **5** shows one of the pair of terminals **16** and the vicinity thereof in the same cross section as FIG. **4**. Thus, FIG. **5** shows a cross section perpendicular or substantially perpendicular to the lateral direction of the battery **1** and passing through the terminal **16**. The other of the pair of terminals **16** and the mounting portion of the terminal **16** to the exterior part **3** (lid member **6**) have the same configuration as described below. That is, the other of the pair of terminals **16** and the insulator **26** and the gasket **27** corresponding to the terminal **16** have the same configuration as that shown in FIG. **5** and the like, and have the same configuration as that described below.

[0036] As shown in FIG. **5** and the like, the terminal **16** has a central axis **C**. In the terminal **16** and the mounting portion of the terminal **16**, a direction along the central axis **C** is defined as an axial direction, and a direction around the central axis **C** is defined as a circumferential direction. In the terminal **16** and the mounting portion of the terminal **16**, a direction orthogonal or substantially orthogonal to the central axis **C** is defined as a radial direction. In the terminal **16** and the mounting portion of the terminal **16**, a side away from the central axis **C** in the radial direction is an outer peripheral side, and a side toward the central axis **C** in the radial direction is an inner peripheral side.

[0037] The terminal **16** includes a terminal shaft **31** and a terminal head **32**. The terminal head **32** is provided adjacent to one side of the terminal shaft **31** in the axial direction of the terminal **16**, and the terminal head **32** protrudes to the outer peripheral side with respect to the terminal shaft **31**. The terminal head **32** protrudes from the terminal shaft **31** to the outer peripheral side over the entire circumference of the terminal **16** in the circumferential direction (direction around the central axis **C**). Therefore, in the terminal **16**, the cross-sectional area of the terminal head **32** in a cross section orthogonal or substantially orthogonal to the central axis **C** is greater than the cross-sectional area of the terminal shaft **31** in a cross section orthogonal or substantially orthogonal to the central axis **C**. In the terminal **16**, an end **E1** on one side in the axial direction is formed by the shaft **31**, and an end **E2** on the opposite side to the end **E1** in the axial direction is formed by the terminal head **32**. The terminal **16** is attached to the lid member **6** in a state where the central axis **C** is along the height direction of the battery **1**, that is, in a state where the axial direction coincides or substantially coincides with the height direction of the battery **1**.

[0038] In the terminal **16**, the terminal shaft **31** is inserted into the internal cavity **10** from the through hole **25** of the lid member **6**. That is, the terminal shaft **31** of the positive electrode terminal **16A** is inserted into the internal cavity **10** from the positive electrode side through hole, and the terminal shaft **31** of the negative electrode terminal **16B** is inserted into the internal cavity **10** from the negative electrode side through hole. In the internal cavity **10**, the terminal shaft **31** of the

terminal **16** penetrates the electrode group holder **21**. In the internal cavity **10**, the terminal shaft **31** of the terminal **16** is connected to the leads **20** (corresponding one of the positive electrode side lead **20A** and the negative electrode side lead **20B**) between the electrode group **2** and the electrode group holder **21** in the height direction of the battery **1**.

[0039] The terminal head **32** of the terminal **16** is arranged outside the exterior part **3**. In the example in FIGS. **1** to **5**, the terminal head **32** is arranged outside the exterior part **3** in a state of protruding from the outer surface of the lid member **6**. As shown in FIG. **5** and the like, the terminal head **32** of the terminal **16** includes head end faces **33** and **35**. In the terminal **16**, the end **E2** is formed by the head end face (first head end face) **33** which is a head upper end face, and in the terminal head **32**, the end on a side far from the terminal shaft **31** in the axial direction is formed by the head end face **33**. In the terminal head **32** of the terminal **16**, an end on the side close to the terminal shaft **31** in the axial direction is formed by the head end face (second head end face) **35** which is a head lower end face, and the head end face **35** is located in a boundary portion with the terminal shaft **31**.

[0040] In the terminal **16**, as described above, the terminal head **32** protrudes to the outer peripheral side with respect to the terminal shaft **31**. Therefore, in the terminal **16**, a step is formed in the boundary portion between the terminal head **32** and the terminal shaft **31** by the head end face **35**. In the terminal **16**, the outer peripheral surface of the terminal head **32** is located toward the outer peripheral side with respect to the outer peripheral surface of the terminal shaft **31** due to the step in the boundary portion between the terminal shaft **31** and the terminal head **32**. In the terminal head **32**, the outer peripheral surface of the terminal head **32** is continuously extended between the head end faces **33** and **35** along the axial direction of the terminal **16**, and is extended over the entire circumference in the circumferential direction of the terminal **16**.

[0041] In the terminal head **32** of the terminal **16**, the end on the side far from the outer surface of the lid member **6** (the outer surface of the exterior part **3**) is formed by the head end face (first head end face) **33**, and the end on the side close to the outer surface of the lid member **6** is formed by the head end face (second head end face) **35**. In the terminal head **32**, the head end face **33** faces the side opposite to the side where the outer surface of the lid member **6** is located, and the head end face **35** opposes the outer surface of the lid member **6**. The terminal head **32** is arranged in a state where the terminal head **32** does not abut on the exterior part **3** (lid member **6**) and a gap is provided between the head end face **35** and the outer surface of the lid member **6**. Thus, a separation distance **D1** between the outer surface of the lid member **6** and the terminal head **32** is defined in the terminal **16**.

[0042] In the terminal head **32** of the terminal **16**, the head end face **33** forms a protruding end of the protruding portion from the outer surface of the lid member **6**, and the head end face **33** serves as a protruding end face of the protruding portion from the outer surface of the lid member **6**. In the terminal **16**, the protruding height **H1** is defined as the dimension between the base position of the protruding portion of the terminal head **32** and the protruding end of the terminal head **32**. In the terminal **16**, the protruding height **H1** corresponds to the distance from the outer surface of the lid member **6** to the head end face **33**, which is a protruding end face, along the axial direction (the height direction of the battery **1**). In the terminal **16**, a distance **D2** corresponding to a distance between the head end faces **33** and **35** of the terminal head **32** along the axial direction is also defined.

[0043] In the example shown in FIGS. **1** to **5**, in the terminal **16**, a distance **D3** from the outer peripheral surface of the terminal head **32** to the closer one of the pair of short edges **17** of the lid member **6** along the lateral direction of the battery **1** is also defined. In the terminal **16**, a distance **D4** from the outer peripheral surface of the terminal head **32** to the pair of long edges **18** of the lid member **6** along the depth direction of the battery **1** is defined. The distance **D4** is smaller than the distance **D3**.

[0044] The gasket **27** covers the terminal shaft **31** of the terminal **16** from the outer peripheral side

in the through hole 25 (corresponding one of the positive electrode side through hole and the negative electrode side through hole). The gasket 27 covers the terminal shaft 31 from the outer peripheral side over the entire circumference in the circumferential direction of the terminal 16. The gasket 27 is sandwiched between the terminal shaft 31 of the terminal 16 and the peripheral edge of the through hole 25. That is, the positive electrode side gasket is sandwiched between the terminal shaft 31 of the positive electrode terminal 16A and the peripheral edge of the positive electrode side through hole, and the negative electrode side gasket is sandwiched between the terminal shaft 31 of the negative electrode terminal 16B and the peripheral edge of the negative electrode side through hole.

[0045] With the above-described configuration, the gasket 27 prevents the terminal 16 from coming into contact with the peripheral edge of the through hole 25. The gasket 27 maintains the space between the peripheral edge of the through hole 25 and the terminal shaft 31 of the terminal 16 in an air-tight and liquid-tight manner. Therefore, in the portion where the terminal 16 is mounted to the exterior part 3, the gasket 27 prevents the electrolyte, gas, and the like from leaking from the internal cavity 10 to the outside of the exterior part 3 through the through hole 25.

[0046] As shown in FIG. 5 and the like, the insulator 26 includes a bottom portion 41 serving as a bottom wall and a covering portion 42 serving as a peripheral wall. In the mounting portion of the terminal 16 to the lid member 6, the insulator 26 is arranged on the outer surface of the lid member 6 (the outer surface of the exterior part 3). In the insulator 26, the bottom portion 41 and the covering portion 42 abut on the outer surface of the lid member 6. The bottom portion 41 of the insulator 26 is adjacent to the gasket 27 from the outer peripheral side in the mounting portion of the terminal 16, and covers the gasket 27 from the outer peripheral side over the entire circumference. The bottom portion 41 of the insulator 26 is arranged in a gap between the terminal head 32 (head end face 35) of the terminal 16 and the outer surface of the lid member 6, and is sandwiched between the terminal head 32 of the terminal 16 and the outer surface of the lid member 6. The terminal 16 penetrates the bottom portion 41 of the insulator 26 and is inserted into the internal cavity 10.

[0047] In the insulator 26, the outer peripheral end of the bottom portion 41 is connected to the covering portion 42, and the bottom portion 41 protrudes from the covering portion 42 to the inner peripheral side over the entire circumference. Therefore, in the mounting portion of the terminal 16 to the lid member 6, the covering portion 42, the bottom portion 41, the gasket 27, and the terminal shaft 31 are arranged in this order from the outer peripheral side. In the insulator 26, the covering portion 42 is extended from a position connected to the bottom portion 41 along the axial direction (central axis C) of the terminal 16. The insulator 26 is arranged in a state where the covering portion 42 protrudes from the outer surface of the lid member 6.

[0048] The covering portion 42 of the insulator 26 covers the terminal head 32 of the terminal 16 from the outer peripheral side on the outer surface of the lid member 6. That is, the positive electrode side insulator 26A covers the terminal head 32 of the positive electrode terminal 16A from the outer peripheral side, and the negative electrode side insulator 26B covers the terminal head 32 of the negative electrode terminal 16B from the outer peripheral side. The insulator 26 covers the terminal head 32 of the terminal 16 from the outer peripheral side over the entire circumference in the circumferential direction. Therefore, in the mounting portion of the terminal 16 to the lid member 6, the covering portion 42 and the terminal head 32 are arranged in this order from the outer peripheral side. In the mounting portion of the terminal 16, the inner peripheral surface of the covering portion 42 of the insulator 26 abuts on the outer peripheral surface of the terminal head 32.

[0049] The covering portion 42 of the insulator 26 includes an extending end face 43. In the covering portion 42, an end on the side opposite to the side connected to the bottom portion 41 is formed by the extending end face 43. The covering portion 42 of the insulator 26 is extended along the axial direction of the terminal 16 from the position connected to the bottom portion 41 to the

extending end face **43**. In the covering portion **42** of the insulator **26**, the end on the side far from the outer surface of the lid member **6** (the outer surface of the exterior part **3**) is formed by the extending end face **43**. In the covering portion **42**, the extending end face **43** faces the side opposite to the side where the outer surface of the lid member **6** is located.

[0050] In the covering portion **42** of the insulator **26**, the extending end face **43** forms the protruding end of the protruding portion from the outer surface of the lid member **6**, and the extending end face **43** serves as a protruding end surface of the protruding portion from the outer surface of the lid member **6**. In the insulator **26**, a protruding height **H2** is defined as a dimension from a base position to a protruding end in the protruding portion of the covering portion **42**. The protruding height **H2** of the insulator **26** corresponds to the distance from the outer surface of the lid member **6** to the extending end face **43**, which is the protruding end face, along the axial direction of the terminal **16** (the height direction of the battery **1**). The protruding height **H2** of the covering portion **42** of the insulator **26** is greater than the distance **D2** between the head end faces **33** and **35** of the terminal head **32** of the terminal **16**. That is, the distance between the extending end face **43** of the covering portion **42** of the insulator **26** and the outer surface of the lid member **6**, which corresponds to the protruding height **H2**, is greater than the distance **D2** between the head end faces **33** and **35**.

[0051] In the present embodiment, in the mounting portion of the terminal **16** to the exterior part **3**, the protruding height **H2** of the covering portion **42** of the insulator **26** from the outer surface of the lid member **6** (the exterior part **3**) is the protruding height **H1** of the terminal head **32** of the terminal **16** from the outer surface of the lid member **6** or more over the entire circumference in the circumferential direction of the terminal **16** (the covering portion **42**). In the example in FIGS. **1** to **5** and the like, the protruding height **H2** of the covering portion **42** is higher than the protruding height **H1** of the terminal head **32** over the entire circumference in the circumferential direction. Therefore, in the example of FIGS. **1** to **5** and the like, the covering portion **42** further protrudes from the head end face **33**, which is the protruding end face of the terminal head **32**, to the side opposite to the side where the outer surface of the lid member **6** is located, over the entire circumference in the circumferential direction.

[0052] In the configuration in which the protruding height **H2** of the covering portion **42** is higher than the protruding height **H1** of the terminal head **32** over the entire circumference in the circumferential direction, as in the example in FIGS. **1** to **5** and the like, a recess **46** is formed with the terminal head **32** of the terminal **16** and the covering portion **42** of the insulator **26**. The recess **46** is recessed to the side opposite to the side to which the terminal head **32** and the covering portion **42** protrude from the outer surface of the lid member **6**. In the recess **46**, a bottom surface serving as a bottom portion and an outer peripheral surface enveloping the recess **46** from the outer peripheral side are defined. The bottom surface of the recess **46** is formed by the terminal head **32** of the terminal **16**, and is formed by the head end face (first head end face) **33** which is a protruding end face of the terminal head **32**. The outer peripheral surface of the recess **46** is formed by the inner peripheral surface of the covering portion **42** and by a portion of the covering portion **42** that further protrudes from the head end face **33** of the terminal head **32**.

[0053] In the covering portion **42** of the insulator **26**, a thickness **T1** is defined as a dimension from the inner peripheral surface to the outer peripheral surface in the radial direction. In one example of the present embodiment, the thickness **T1** of the covering portion **42** of the insulator **26** is the separation distance **D1** between the outer surface of the lid member **6** and the terminal head **32** or more. The thickness **T1** of the covering portion **42** is the separation distance **D1** or more over the entire circumference in the circumferential direction. Therefore, the covering portion **42** is formed to the thickness **T1** having a thickness of the gap between the terminal head **32** and the outer surface of the lid member **6** or more.

[0054] However, in the present embodiment, in the projected view from the height direction of the battery **1**, the outer peripheral surface of the covering portion **42** is positioned in a range

surrounded by the peripheral edge **14** of the lid member **6** over the entire circumference in the circumferential direction. The entire insulator **26** including the covering portion **42** is located between the pair of short edges **17** in the lateral direction of the battery **1** and between the pair of long edges **18** in the depth direction of the battery **1**. Therefore, the covering portion **42** does not protrude outward beyond the peripheral edge **14** of the lid member **6**, and does not protrude outward in the depth direction of the battery **1** beyond each of the long edges **18**.

[0055] As shown in FIG. **3** and the like, the covering portion **42** of the insulator **26** that covers the terminal head **32** of the terminal **16** from the outer peripheral side includes two pairs of adjacent portions **51** and **52**. The pair of adjacent portions (first adjacent portions) **51** are adjacent to the terminal head **32** from opposite sides to each other in the lateral direction of the battery **1**.

Therefore, the terminal head **32** is sandwiched between the pair of adjacent portions **51** in the lateral direction of the battery **1**. The pair of adjacent portions (second adjacent portions) **52** are adjacent to the terminal head **32** from opposite sides to each other in the depth direction of the battery **1**. Therefore, the terminal head **32** is sandwiched between the pair of adjacent portions **52** in the depth direction of the battery **1**.

[0056] In the present embodiment, the dimension of the lid member **6** in the depth direction of the battery **1** is smaller than the dimension of the lid member **6** in the lateral direction of the battery **1**, and the distance **D4** is smaller than the distance **D3**. Thus, in the present embodiment, it is preferable that a thickness **T1b** of each of the adjacent portions (second adjacent portions) **52** is smaller than a thickness **T1a** of each of the adjacent portions (first adjacent portions) **51**.

[0057] As shown in FIG. **5** and the like, in the present embodiment, a pair of recessed portions **55** are formed in the outer surface of the lid member **6**, and one recessed portion **55** is formed in each of the mounting portions of the terminals **16** to the lid member **6**. In the projected view from the height direction of the battery **1**, the recessed portions **55** are located in a range surrounded by the peripheral edge **14** of the lid member **6**. Hereinafter, one of the pair of recessed portions **55** and a configuration related to the recessed portion **55** will be described. The other of the pair of recessed portions **55** and the configuration related to the recessed portion **55** are the same as the configuration described below.

[0058] In the mounting portion of the terminal **16**, the recessed portion **55** is recessed toward the side opposite to the side where the terminal head **32** and the covering portion **42** protrude from the outer surface of the lid member **6**. The recessed portion **55** has a recess bottom surface **56** as a bottom portion and a recess outer peripheral surface **57** enveloping the recessed portion **55** from the outer peripheral side. In the mounting portion of the terminal **16**, the through hole **25** is formed in the recess bottom surface **56** of the recessed portion **55**. In the terminal **16**, the entire terminal head **32** is located on the inner peripheral side of the recessed portion **55** with respect to the recess outer peripheral surface **57**. Therefore, the outer peripheral surface of the terminal head **32** is located on the inner peripheral side of the recessed portion **55** (terminal **16**) with respect to the recess outer peripheral surface **57**. The terminal head **13** of the terminal **16** sandwiches the bottom portion **41** of the insulator **26** between the head end face **35** and the recess bottom surface **56**.

[0059] The covering portion **42** of the insulator **26** is extended from a position on the inner peripheral side with respect to the recess outer peripheral surface **57** toward the outer peripheral side of the recessed portion **55**. The covering portion **42** extends beyond the recess outer peripheral surface **57** to a position on the outer peripheral side with respect to the recess outer peripheral surface **57**. Thus, the covering portion **42** includes an inner peripheral portion **61** on the inner peripheral side with respect to the recess outer peripheral surface **57** and an outer peripheral portion **62** on the outer peripheral side with respect to the recess outer peripheral surface **57**. In the covering portion **42**, each of the inner peripheral portion **61** and the outer peripheral portion **62** is formed over the entire circumference in the circumferential direction. In the covering portion **42**, the inner peripheral portion **61** abuts on the recess bottom surface **56**, and the outer peripheral portion **62** abuts on the outer surface of the lid member **6** (the outer surface of the exterior part **3**) in

a region on the outer peripheral side with respect to the recess outer peripheral surface 57.

[0060] In the covering portion 42, a dimension T2 occupied by the inner peripheral portion 61 in the thickness T1 of the covering portion 42 is defined as the dimension of the inner peripheral portion 61 in the radial direction of the terminal 16. In the covering portion 42, a dimension T3 occupied by the outer peripheral portion 62 in the thickness T1 of the covering portion 42 is defined as the dimension of the outer peripheral portion 62 in the radial direction of the terminal 16. In the covering portion 42, the dimension T3 is the dimension T2 or more over the entire circumference in the circumferential direction. The dimension T3 corresponds to the protruding length of the portion of the covering portion 42 protruding from the recessed portion 55 to the outer peripheral side.

[0061] Further, a distance D5 from the recess outer peripheral surface 57 to the respective long edges 18 of the lid member 6 along the depth direction of the battery 1 is defined. In the present embodiment, as described above, the entire insulator 26 including the covering portion 42 is located between the pair of long edges 18 in the depth direction of the battery 1. Therefore, the dimension T3 occupied by the outer peripheral portion 62 in the thickness T1 of the covering portion 42 is smaller than the distance D5.

[0062] In the present embodiment, a battery module is assembled using a plurality of batteries 1 as described above. FIG. 6 shows an example of a battery module 70 of the present embodiment. As shown in FIG. 6, the battery module 70 includes a plurality of the batteries 1 described above, and the plurality of batteries 1 are aligned. In the battery module 70, a depth direction (a direction indicated by an arrow X3 and an arrow X4), a lateral direction (a direction indicated by an arrow Y3 and an arrow Y4) intersecting (orthogonal or substantially orthogonal) with the depth direction, and a height direction (a direction indicated by an arrow Z3 and an arrow Z4) intersecting (orthogonal or substantially orthogonal) with both the depth direction and the lateral direction are defined. In the battery module 70, the plurality of batteries 1 are aligned along the depth direction, and the depth direction coincides or substantially coincides with the alignment direction of the batteries 1.

[0063] Each of the plurality of batteries 1 is arranged in a state where the depth direction coincides or substantially coincides with the depth direction of the battery module 70, and the lateral direction coincides or substantially coincides with the lateral direction of the battery module 70. The height direction of each of the plurality of batteries 1 coincides or substantially coincides with the height direction of the battery module 70. In the battery module 70, the outer surfaces of the lid members 6 of the plurality of batteries 1 face the same side as each other in the height direction. In the battery module 70, the terminal heads 32 of the terminals 16 of the plurality of batteries 1 protrude from the outer surface of the lid member 6 toward the same side as each other in the height direction.

[0064] In the battery module 70, the plurality of batteries 1 are electrically connected to each other with the bus bars 71 formed of a conductive material. Therefore, one of the plurality of batteries 1 is electrically connected to an additional battery, which is another battery 1, via the bus bar 71. In the example of FIG. 6, the batteries 1 adjacent to each other in the alignment direction are electrically connected in series by one bus bar 71. The plurality of batteries 1 provided in the battery module 70 are electrically connected in series by using the plurality of bus bars 71. In the battery module 70, the plurality of batteries 1 may be electrically connected in parallel by using one or more bus bars 71. Further, both a structure in which the plurality of batteries 1 are electrically connected in series and a structure in which the plurality of batteries 1 are electrically connected in parallel may be formed by one or more bus bars.

[0065] The bus bar 71 is formed in a plate shape, and a plate thickness direction is defined in the bus bar 71. In the battery module 70, the bus bar 71 electrically connects the two batteries 1 in a state where the plate thickness direction of the bus bar 71 coincides or substantially coincides with the height direction of the battery module 70. Further, the bus bar 71 is formed with an insertion

piece part **72**. In the bus bar **71**, the insertion piece part **72** protrudes toward one side in the plate thickness direction. In each of the plurality of batteries **1** of the battery module **70**, the protruding height **H2** of the covering portion **42** of the insulator **26** is higher than the protruding height **H1** of the terminal head **32** of the terminals **16** over the entire circumference in the circumferential direction in the mounting portion of each of the pair of terminals **16**. Therefore, in each of the plurality of batteries **1**, the above-described recess **46** is formed in the mounting portion of each of the pair of terminals **16**.

[0066] FIG. **7** illustrates a configuration in which bus bars **71** are connected to the terminals **16** of the batteries **1** in the battery module **70**. In FIG. **7**, a cross section orthogonal or substantially orthogonal to the lateral direction of the battery module **70** is shown. In addition, FIG. **7** shows a cross section passing through one of the pair of terminals **16** in each of the plurality of batteries **1** and one of the bus bars **71**. Hereinafter, a configuration in which the bus bar **71** is connected to any one of the pair of terminals **16** in one of the plurality of batteries **1** will be described. In the battery module **70**, the other terminals **16** are also connected to the bus bars **71** in the same manner as in the configuration described below.

[0067] In the present embodiment, the insertion piece part **72** of the bus bar **71** is inserted into the recess **46** formed by the terminal head **32** and the covering portion **42**. The insertion piece part **72** is joined to the terminal head **32** in a state of being inserted into the recess **46**. That is, in the recess **46**, the insertion piece part **72** is joined to the head end face (first head end face) **33** of the terminal head **32**, which serves as the bottom surface of the recess **46**. The insertion piece part **72** is joined to the terminal head **32** in the recess **46**, and thus the bus bar **71** is connected to the terminal **16**.

[0068] In the recess **46**, the insertion piece part **72** of the bus bar **71** abuts on the outer peripheral surface of the recess **46**, that is, the inner peripheral surface of the covering portion **42**. In the present embodiment, in the projected view from the height direction of the battery **1**, the recess **46** has a quadrangular shape or a substantially quadrangular shape. In addition, in the projected view from the plate thickness direction of the bus bar **71**, the insertion piece part **72** has a quadrangular shape or a substantially quadrangular shape which is similar to the quadrangular shape or the substantially quadrangular shape of the recess **46**, and has a slightly smaller area than the quadrangular shape or the substantially quadrangular shape of the recess **46**. In the recess **46**, the insertion piece part **72** abuts on the inner peripheral surface of the covering portion **42** over the entire circumference or substantially the entire circumference of the covering portion **42** (terminal **16**) in the circumferential direction.

[0069] In the case of assembling a battery module **70** or the like in which a plurality of batteries **1** are electrically connected, a bus bar **71** is connected to the terminal **16**, and the plurality of batteries are electrically connected via the bus bar **71**. In addition, to confirm the dimensions of the battery module **70** to be assembled, measurement and the like of the dimensions of the batteries **1** forming the battery module **70** are performed. In the battery **1** of the present embodiment, the terminal head **32** is covered by the covering portion **42** of the insulator **26** from the outer peripheral side in the respective mounting portions of the terminals **16**. The covering portion **42** abuts on the outer surface of the exterior part **3** (the outer surface of the lid member **6**), and the protruding height **H2** of the covering portion **42** from the outer surface of the exterior part **3** is the protruding height **H1** of the terminal head **32** from the outer surface of the exterior part **3** or more over the entire circumference.

[0070] With such a configuration, in the work of assembling the battery module **70** or the like, the covering portion **42** appropriately prevents the terminal head **32** from unintentionally coming into contact with a conductor such as a metal component or a jig. Thus, in the work of assembling the battery module using the batteries **1** in which the terminal heads **32** of the terminals **16** are arranged outside the exterior parts **3**, the occurrence of a short circuit caused by contact of the terminal heads **32** with the conductors is appropriately prevented.

[0071] In addition, since the protruding height **H2** of the covering portion **42** is the protruding

height H1 of the terminal head 32 or more in the respective mounting portions of the terminals 16, the clearance and the creepage distance of the outer surface of the exterior parts 3 (outer surface of the lid members 6) of the batteries 1 with respect to the bus bars 71 are increased in the battery module 70. This further appropriately prevents the occurrence of a short circuit between the bus bar 71 and the exterior part 3.

[0072] In the present embodiment, the protruding height H2 of the covering portion 42 is set to be higher than the protruding height H1 of the terminal head 32 over the entire circumference, and thus the recess 46 is formed in which the head end face (first head end face) 33 of the terminal head 32 is recessed with respect to the extending end face 43 of the covering portion 42. The formation of the recess 46 further appropriately prevents the terminal head 32 from unintentionally contacting the conductor. Further, with the configuration in which the recess 46 is formed in each of the mounting portions of the terminals 16, the creepage distance between the terminal head 32 and the region on the outer peripheral side with respect to the covering portion 42 of the outer surface of the exterior part 3 increases. Thus, in the battery 1, the occurrence of a short circuit between each of the terminals 16 and the exterior part 3 is further appropriately prevented.

[0073] As described above, in the present embodiment, the insulation between the conductors is improved in the battery 1 and the battery module 70 in which the plurality of batteries 1 are assembled. Therefore, at the time of using the battery 1 and the like, the occurrence of a short circuit is appropriately prevented, and safety is improved. In addition, the occurrence of a short circuit is appropriately prevented during the assembly work of the battery module 70, during the use of the battery module 70, and the like, and thus safety is improved.

[0074] In the present embodiment, the distance between the extending end face 43 of the covering portion 42 of the insulator 26 and the outer surface of the exterior part 3 is greater than the distance D2 between the head end faces 33 and 35 of the terminal head 32. Therefore, the configuration in which the protruding height H2 of the covering portion 42 of the insulator 26 is the protruding height H1 of the terminal head 32 of the terminal 16 or more in the mounting portion of each of the terminals 16 can be easily realized.

[0075] In the example of the battery 1 shown in FIGS. 1 to 5, the thickness T1 of the covering portion 42 of the insulator 26 is the separation distance D1 between the outer surface of the exterior part 3 and the terminal head 32 or more. In the example of FIGS. 1 to 5, the recessed portion 55 is formed on the outer surface of the exterior part 3 at each of the mounting portions of the terminals 16, and the terminal head 32 is located on the inner peripheral side with respect to the recess outer peripheral surface 57 of the recessed portion 55. The covering portion 42 of the insulator 26 extends from a position on the inner peripheral side with respect to the recess peripheral surface 57 to the outer peripheral side of the recessed portion 55 beyond the recess outer peripheral surface 57. With such a configuration, the thickness T1 of the covering portion 42 increases. In addition, in the case where the recessed portion 55 is formed, the thickness T1 of the covering portion 42 is further increased by setting the dimension T3 occupied by the outer peripheral portion 62 on the outer peripheral side with respect to the recess outer peripheral surface 57 to be the dimension T2 occupied by the inner peripheral portion 61 on the inner peripheral side with respect to the recess outer peripheral surface 57 or more.

[0076] In each of the batteries 1 forming the battery module 70, the thickness T1 of the covering portion 42 increases in the mounting portion of each of the terminals 16, and thus the movement of the bus bar 71 connected to the terminals 16 is appropriately suppressed by the covering portion 42. For example, even if a force acts on the covering portion 42 from the bus bar 71 due to vibration or the like of the battery module 70, a reaction force against the force from the bus bar 71 appropriately acts on the bus bar 71 from the covering portion 42, and the movement of the bus bar 71 is appropriately prevented by the reaction force from the covering portion 42. That is, the fixing force of the bus bar 71 connected to the terminal 16 is improved. This effectively prevents the bus bar 71 from being separated from the terminal 16 due to vibration or the like of the battery module

70, and improves the structural reliability of the battery module **70**.

[0077] In the battery module **70** of the example shown in FIGS. **6** and **7** and the like, the insertion piece part **72** of the bus bar **71** is inserted into the recess **46** formed by the terminal head **32** and the covering portion **42** in the connection portion of the bus bar **71** to each of the terminals **16**. The insertion piece part **72** of the bus bar **71** is joined to the terminal head **32** in the recess **46**. With such a configuration, the bus bar **71** can be easily positioned during assembling of the battery module **70**.

[0078] In the battery module **70**, the insertion piece part **72** of the bus bar **71** abuts on the outer peripheral surface of the recess **46**, that is, the covering portion **42**, in the recess **46**. This makes it easier to position the bus bar **71** during assembling of the battery module **70**. Since the bus bar **71** is easily positioned in the assembly of the battery module **70**, the workability of the work of assembling the battery module **70** is improved. This improves the productivity of the battery module **70**.

[0079] In the battery **1** of the example shown in FIGS. **1** to **5**, in the mounting portion of each terminal **16**, the covering portion **42** includes a pair of adjacent portions **51** adjacent to the terminal head **32** from opposite sides to each other in the lateral direction, and a pair of adjacent portions **52** adjacent to the terminal head **32** from opposite sides to each other in the depth direction. In one example, the thickness **T1b** of each of the adjacent portions **52** is smaller than the thickness **T1a** of each of the adjacent portions **51**. In the manufacturing of the battery **1**, the lid member **6** is welded to the exterior container **5**, as described above. By making the thickness **T1b** thinner by making the thickness **T1b** smaller than the thickness **T1a** and the like, the influence of heat, generated by welding the lid member **6** to the exterior container **5**, on the covering portion **42** is reduced.

(Modification)

[0080] In the connection portion of the bus bar **71** to each of the terminals **16**, the insertion piece part **72** of the bus bar **71** may abut on the inner peripheral surface of the covering portion **42** (the outer peripheral surface of the recess **46**) only in a partial range in the circumferential direction of the covering portion **42** (the terminal **16**). In a battery module **70** of a first modification shown in FIG. **8**, in the projected view from the plate thickness direction of the bus bar **71**, the insertion piece part **72** has an elliptical shape or a substantially elliptical shape having a smaller area than the quadrangular shape or the substantially quadrangular shape of the recess **46**. In the battery module **70**, the bus bar **71** electrically connects the two batteries **1** in a state where the major axis direction of the elliptical shape or the substantially elliptical shape of the insertion piece part **72** coincides or substantially coincides with the depth direction of the battery module **70** (the depth direction of each of the batteries **1**).

[0081] In this modification, the insertion piece part **72** of the bus bar **71** is inserted into the recess **46** in the connection portion of the bus bar **71** to each of the terminals **16**, and the insertion piece part **72** is joined to the head end face **33** of the terminal head **32** in the recess **46**. In this modification, the insertion piece part **72** abuts on the inner peripheral surface of the covering portion **42** (the outer peripheral surface of the recess **46**) only at two locations in the recess **46** that are spaced apart from each other in the circumferential direction of the terminal **16** (the covering portion **42**). In the inner peripheral surface of the covering portion **42**, a portion on which the insertion piece part **72** abuts is formed in each of the pair of adjacent portions (second adjacent portions) **52**, and the insertion piece part **72** does not abut on the pair of adjacent portions (first adjacent portions) **51**. In the recess **46**, two locations where the insertion piece part **72** abuts on the covering portion **42** are located at an interval of, for example, 180° or substantially 180° in the circumferential direction of the terminal **16**.

[0082] In a modification, the bus bar **71** electrically connects two batteries **1** in a state where the major axis direction of the elliptical shape or the substantially elliptical shape of the insertion piece part **72** coincides or substantially coincides with the lateral direction of the battery module **70** (the lateral direction of each of the batteries **1**). Also, in this case, the insertion piece part **72** abuts on

the inner peripheral surface of the covering portion **42** (the outer peripheral surface of the recess **46**) only in two locations separated from each other in the circumferential direction of the terminal **16** (the covering portion **42**) in the recess **46**. However, in this modification, on the inner peripheral surface of the covering portion **42**, a portion on which the insertion piece part **72** abuts is formed at each of the pair of adjacent portions (first adjacent portions) **51**, and the insertion piece part **72** does not abut on the pair of adjacent portions (second adjacent portions) **52**.

[0083] In a battery module **70** of a second modification shown in FIG. **9**, in the projected view from the plate thickness direction of the bus bar **71**, the insertion piece part **72** has a circular shape or a substantially circular shape having a smaller area than the quadrangular shape or the substantially quadrangular shape of the recess **46**. In this modification, the insertion piece part **72** abuts on the inner peripheral surface of the covering portion **42** (the outer peripheral surface of the recess **46**) only at four locations spaced apart from each other in the circumferential direction of the terminal **16** (the covering portion **42**) in the recess **46**. In the inner peripheral surface of the covering portion **42**, a portion on which the insertion piece part **72** abuts is formed at one location each on the pair of adjacent portions (first adjacent portions) **51** and the pair of adjacent portions (second adjacent portions) **52**. In the recess **46**, four locations on which the insertion piece parts **72** abut the covering portion **42** are located at intervals of, for example, 90° or substantially 90° in the circumferential direction of the terminal **16**.

[0084] In the modification example of FIG. **8** and the modification example of FIG. **9** and the like, the insertion piece part **72** of the bus bar **71** is joined to the terminal head **32** in the recess **46**, as in the first embodiment and the like. In addition, the insertion piece part **72** of the bus bar **71** abuts on the outer peripheral surface of the recess **46**, that is, the covering portion **42**, in the recess **46**. Therefore, in the above-described modification and the like, the bus bar **71** can be easily positioned during assembling of the battery module **70**. Therefore, in the above-described modifications and the like, the workability of the work of assembling the battery module **70** is improved, and the productivity of the battery module **70** is improved.

[0085] In the above-described embodiment and the like, the exterior part **3** includes the exterior container **5** and the lid member **6**, but the configuration of the exterior part is not limited to the above. As long as the pair of terminals **16** are attached to the exterior part, the same configuration as that of any of the above-described embodiments and the like can be applied to the mounting portions of the terminals **16** to the exterior part.

[0086] That is, in the above-described embodiment, modifications, and the like, the terminal head **32** of the terminal **16** is arranged outside the exterior part in the mounting portion of each terminal **16** to the exterior part. The bottom portion **41** of the insulator **26** is sandwiched between the terminal head **32** and the outer surface of the exterior part, and the covering portion **42** of the insulator **26** covers the terminal head **32** from the outer peripheral side of the terminal **16** on the outer surface of the exterior part. The covering portion **42** abuts on the outer surface of the exterior part, and the protruding height **H2** of the covering portion **42** from the outer surface of the exterior part is the protruding height **H1** of the terminal head **32** from the outer surface of the exterior part or more over the entire circumference. With such a configuration, in the work of assembling the battery module and the like, the covering portion **42** appropriately prevents the terminal head **32** from unintentionally coming into contact with the conductor. In the battery **1** and the battery module in which the plurality of batteries **1** are assembled, the insulation between the conductors is improved, and the safety is improved.

[0087] In the battery **1**, at least one of the mounting portion of the positive electrode terminals **16A** to the exterior part and the mounting portion of the negative electrode terminals **16B** to the exterior part may have the same configuration as that of any of the above-described embodiments and the like. In the case where the mounting portion of the positive electrode terminals **16A** to the exterior part has the same configuration as that of the above-described embodiments and the like, the positive electrode terminals **16A** have the same configuration as that of any of the above-described

terminals **16**, and the positive electrode side insulators **26A** have the same configuration as that of any of the above-described insulators **26**. In the case where the mounting locations of the negative electrode terminals **16B** to the exterior part are the same positions as those in the above-described embodiments and the like, the negative electrode terminals **16B** have the same configuration as any of the terminals **16** described above, and the negative electrode side insulators **26B** have the same configuration as any of the insulators **26** described above.

[0088] According to at least one of these embodiments or examples, the bottom portion of the insulator is sandwiched between the terminal head and the outer surface of the exterior part, and the covering portion of the insulator covers the terminal head from the outer peripheral side of the terminal on the outer surface of the exterior part. The covering portion abuts on the outer surface of the exterior part, and the protruding height of the covering portion from the outer surface of the exterior part is the protruding height of the terminal head from the outer surface of the exterior part or more over the entire circumference. Thus, it is possible to provide a battery in which unintended contact of the terminal head arranged outside the exterior part with the conductor is appropriately prevented in the work of assembling the battery module and the like, and a battery module including such batteries.

[0089] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A battery comprising: an exterior part having an internal cavity; an electrode group including a positive electrode and a negative electrode, and housed in the internal cavity; a terminal inserted into the internal cavity through the exterior part and electrically connected to the electrode group, the terminal including a terminal head arranged outside the exterior part; and an insulator having an electrical insulation property, the insulator including a bottom portion sandwiched between the terminal head and an outer surface of the exterior part, and a covering portion covering the terminal head from an outer peripheral side of the terminal on the outer surface of the exterior part, the covering portion abutting on the outer surface of the exterior part, a protruding height of the covering portion from the outer surface of the exterior part being a protruding height of the terminal head from the outer surface of the exterior part or more over an entire circumference.
2. The battery according to claim 1, wherein a thickness of the covering portion is a separation distance between the outer surface of the exterior part and the terminal head or more.
3. The battery according to claim 1, wherein the terminal head of the terminal includes: a first head end face that forms an end of the terminal head on a side far from the outer surface of the exterior part, and that faces a side opposite to a side on which the outer surface of the exterior part is located; and a second head end face that forms an end of the terminal head on a side close to the outer surface of the exterior part, and that opposes the outer surface of the exterior part with the bottom portion of the insulator sandwiched between the second head end face and the outer surface of the exterior part, the covering portion of the insulator includes an extending end face that forms an end of the covering portion on the side far from the outer surface of the exterior part, and that faces the side opposite to the side on which the outer surface of the exterior part is located, and a distance between the outer surface of the exterior part and the extending end face of the covering part is greater than a distance between the first head end face and the second head end face of the terminal head.

- 4.** The battery according to claim 1, wherein the exterior part includes: an exterior container including a bottom wall and a peripheral wall, the internal cavity being open to a side opposite to a side where the bottom wall is located in a height direction; and a lid member that is attached to the peripheral wall of the exterior container in a state of closing an opening of the internal cavity, and through which the terminal penetrates, the bottom portion of the insulator is sandwiched between the terminal head and an outer surface of the lid member, and the covering portion of the insulator covers the terminal head from the outer peripheral side of the terminal on the outer surface of the lid member.
- 5.** The battery according to claim 4, wherein a dimension of the lid member in a depth direction intersecting the height direction is smaller than a dimension of the lid member in a lateral direction intersecting both the height direction and the depth direction, the covering portion of the insulator includes a pair of first adjacent portions adjacent to the terminal head of the terminal from opposite sides to each other in the lateral direction, and a pair of second adjacent portions adjacent to the terminal head of the terminal from opposite sides to each other in the depth direction, and a thickness of each of the second adjacent portions is smaller than a thickness of each of the first adjacent portions.
- 6.** The battery according to claim 4, wherein the lid member includes a peripheral edge forming an edge of the outer surface of the lid member, and in a projected view from the height direction, an outer peripheral surface of the covering portion of the insulator is located in a range surrounded by the peripheral edge of the lid member over the entire circumference.
- 7.** The battery according to claim 1, wherein the exterior part includes a recessed portion that is recessed in the outer surface of the exterior part and that defines a recess bottom surface and a recess outer peripheral surface, the terminal head of the terminal is located on an inner peripheral side of the recessed portion with respect to the recess outer peripheral surface, and sandwiches the bottom portion of the insulator between the terminal head and the recess bottom surface, and the covering portion of the insulator is extended from a location on the inner peripheral side of the recessed portion with respect to the recess outer peripheral surface to an outer peripheral side of the recessed portion beyond the recess outer peripheral surface.
- 8.** The battery according to claim 7, wherein in a thickness of the covering portion of the insulator, a dimension occupied by an outer peripheral portion on the outer peripheral side with respect to the recess outer peripheral surface is a dimension occupied by an inner peripheral portion on the inner peripheral side with respect to the recess outer peripheral surface or more.
- 9.** The battery according to claim 1, wherein at least one of following is true: a positive electrode terminal electrically connected to the positive electrode of the electrode group is the terminal including the terminal head, and a positive electrode side insulator corresponding to the positive electrode terminal is the insulator including the bottom portion and the covering portion; and a negative electrode terminal electrically connected to the negative electrode of the electrode group is the terminal including the terminal head, and a negative electrode side insulator corresponding to the negative electrode terminal is the insulator including the bottom portion and the covering portion.
- 10.** A battery module comprising: the battery according to claim 1; a bus bar formed of a conductive material and connected to the terminal of the battery; and an additional battery electrically connected to the battery via the bus bar.
- 11.** The battery module according to claim 10, wherein in the battery, a recess is formed with the terminal head of the terminal and the covering portion of the insulator, in the recess, a bottom surface of the recess is formed with the terminal head, the bus bar includes an insertion piece part to be inserted into the recess, and the insertion piece part of the bus bar is joined to the terminal head of the terminal in the recess.
- 12.** The battery module according to claim 11, wherein in the recess, an outer peripheral surface of

the recess is formed with the covering portion, and the insertion piece part of the bus bar abuts on the outer peripheral surface of the recess in the recess.
